5.1 Parametric Worksheet

1. Sketch the curve of the following parametric function by plotting points. Indicate with arrows the direction in which the curve is traced as t increases.

$$x = 2\cos(t), \qquad y = t - \cos(t), \qquad 0 \le t \le 2\pi$$

2. Consider the parametric curve given by the parametric equations

$$x = 1 + t, \qquad y = 5 - 2t, \qquad -2 \le t \le 3$$

(a) Sketch the curve of the following parametric function by plotting points. Indicate with arrows the direction in which the curve is traced as t increases.

(b) Find a Cartesian equation for this curve.

3. Consider the parametric curve given by the parametric equations

$$x = 4\cos(\theta), \qquad y = 5\sin(\theta), \qquad -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$$

(a) Sketch the curve of the following parametric function by plotting points. Indicate with arrows the direction in which the curve is traced as t increases.

(b) Find a Cartesian equation for this curve.

4. Consider the curve given by the parametric equations

$$x = 2e^{3t} + t^2, \qquad y = 5e^{2t}, \qquad 0 \le t \le 1$$

(a) Find a formula for $\frac{dy}{dx}$.

(b) What is the equation of the line tangent to the curve at the point $(2,5)$?

(c) Set up, but do not evaluate, an integral with respect to t that represent the arc length of the curve.

5. Consider the curve given by the parametric equations

$$x = 3\cos\left(\frac{t}{2}\right), \qquad y = t^2, \qquad 0 \le t \le 4\pi$$

(a) What are the slopes of the tangent lines to the curve at the point when $x = 0$.

(b) Set up, but do not evaluate, an integral with respect to t that represents the arc length of the curve.

(c) Determine the second derivative of the parametric curve give by $x = 3\cos\left(\frac{t}{2}\right), \quad y = t^2, \quad 0 \le t \le 4\pi$.